Ellis Capability Map

What Ellis can do today, the platform-verified stories behind each capability, and what it is explicitly not allowed to do yet.

40 / 48
capability stories verified by the platform, mapped to 39 capabilities

47 / 55
stories verified in all, with discovery, the migration plan and the Command Center

16
higher sandbox rungs built (R3 or R4), each on a sandbox copy only

926
tests passing, each capability's key rule shown to fail when broken

Built is not the same as approved. Every capability here is built and verified by the platform from a commit that names its story. None has an approved autonomy level on record yet: Ellis's own authority check resolves all 48 to none. So today Ellis answers from records and rehearses fixes on sandbox copies. It does not change the live platform.

How to read a page. What it does; a ladder showing how far it is built and where its ceiling is; the verified stories with their acceptance criteria and commits; then what it may not do *yet* (it would need an approved step up) and what it may *never* do, at any level.

THE AUTONOMY LADDER (PLATFORM DEFINITION)

R0	Define the workflow, requirements, risks and human decisions
R1	Prepare isolated development, fixtures and baseline tests
R2	Answer questions using approved knowledge and records
R3	Perform approved, user-initiated sandbox tool operations
R4	Execute and verify stateful, multi-step sandbox tasks
R5	Evaluate in shadow mode against approved reference outcomes
R6+	Supervised production pilot, then authorized automation; each needs its own approval

TEN LIMITS THAT APPLY TO EVERY CAPABILITY

- Never merges code or deploys
- Never writes to the platform's code or data
- Never adopts a prompt change or technology direction
- Never lets a sandbox test reach the network
- Never changes a live schedule, or replays or abandons a dead letter
- Never stops, starts, pauses or resumes a live agent
- Never raises its own autonomy or widens its own policy
- Never changes the AI model class
- Never runs a sandbox operation unattended
- Never creates a missing switch or record

All capabilities at a glance

#	CAPABILITY	BUILT TO	CEILING
HEALTH AND RESILIENCE			
1	Schema drift detection	R2 + R3 sandbox	R3 · human
2	Orchestration health watch	R2	R5
3	System resilience supervision	R2	R5
4	Agent registry read surface	R2	R2
5	Cron run instrumentation	R2	R2
6	Cron health alerting	R2	R5
7	Agent health scoring	R2 + R4 sandbox	R4
8	Timer and agent registry reconciliation	R2 + R4 sandbox	R4
9	Incident trace and error triage	R2	R5
10	Error classification coverage	R2 + R4 sandbox	R4
11	Scheduler registration liveness probe	R2 + R4 sandbox	R4
SECURITY			
12	Static security posture scan	R2	R5
13	Runtime threat and access watch	R2	R5
14	Agent behaviour and AI-safety watch	R2	R6
15	Security oversight rollup	R2	R5
MAINTENANCE AND RECOVERY			
16	Backup and restore verification	R2 + R3 sandbox	R3 · human
17	Deploy verification evidence	R2	R2 · human
18	Orchestration auto-repair (sandbox only)	R4	R6
19	Data retention sweeps	R2 + R4 sandbox	R4
20	Preview stack reaping	R2 + R4 sandbox	R4
21	Stale action recovery	R2 + R4 sandbox	R4
22	Ticket auto-resolution (sandbox only)	R4	R7
23	Bug fix dispatch	R1 + R4 sandbox	R4 · human
24	Dead-letter capture	R2 + R4 sandbox	R4
25	Dead-letter triage	R2 + R4 sandbox	R4 · human
26	Idempotency and replay audit	R1 + R4 sandbox	R4
27	Cron overlap guard	R2 + R4 sandbox	R4
28	Production deploy review	R1	R1 · human
DIRECTION AND CONTROL			
29	Infrastructure evolution proposals	R2	R5 · human
30	Agent enable, disable and pause control	R2	R2 · human
31	Workforce technology direction	R2	R5 · human
32	Prompt and performance optimisation	R2	R5 · human
33	Schedule governance and live reload	R2	R2 · human
34	Default-on timer guard	R2 + R4 sandbox	R4 · human
35	Reliability briefing	R2	R5
36	Reliability threshold ownership	R2	R5 · human
37	Agent promotion evidence	R2	R2 · human
38	Autonomy authority resolution	R2	R2 · human
39	Manager access	R2	R2 · human

Schema drift detection

First release R2 · Sandbox rung R3 built · Ceiling R3 · Human decision at every level · Approved level on record: none

WHAT IT DOES

Reports what the latest nightly drift run found, and names what that run never compares. Where no run has completed, it says the answer is unknown rather than reporting no drift.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX				

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.schema_drift_detection` Where no drift run has completed, the answer is 'unknown: last run not found', never 'no drift'.	Verified 1 Nov 2026	15d81be
`R3 sandbox rung` compares two sandbox databases both ways	Built	53814d0

NOT ALLOWED YET

- Read the production schema. Production database access is denied to agents by the platform.
- Start a comparison on its own. The R3 comparison runs only when a person starts it, between two sandbox databases.

NEVER, AT ANY LEVEL

- Apply a schema change or re-sync development. That stays a person's decision at every level.

Orchestration health watch

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Answers whether orchestration is healthy right now and when that last changed. It reads one agent's registry row and its activity log, and every answer cites the rows it came from.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.orchestration_health_watch` Every answer cites the ai_agents row and activity-log rows it was derived from; no answer is produced for an agent name with no row.	Verified 26 Oct 2026	`cd225af`

NOT ALLOWED YET

- Run in shadow mode (R5) against approved reference outcomes. That waits on K.D. setting acceptable misses and false alarms.
- Raise an alert. It reports what it sees; alerting stays with the platform's own checks.

NEVER, AT ANY LEVEL

- Write to the platform or restart anything. It reads two tables and nothing else.

Write-up: docs/capabilities/orchestration-health-watch.md · Catalog row 1 · github.com/qninying/platform-reliability-agent

System resilience supervision

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Gives each part of the platform a state and the age of its last signal. A part with no signal is unknown and blocks a healthy verdict. It also says what the platform's own health probes report, and what they cannot see, such as a scheduler that never registered.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.system_resilience_supervision` A subsystem with no signal is reported as unknown and prevents a healthy overall verdict.	Verified 27 Oct 2026	9b251a1
`platform.CAP.system_health_probe_and_alerting` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-

NOT ALLOWED YET

- Run in shadow mode (R5). That waits on K.D.'s thresholds.
- Raise an alert from the probes. It discloses their blind spots; it does not replace them.

NEVER, AT ANY LEVEL

- Restart or change any service.

Write-up: docs/capabilities/system-resilience-supervision.md · Catalog row 12 · github.com/qninying/platform-reliability-agent

Agent registry read surface

First release R2 | Ceiling R2 | Approved level on record: none

WHAT IT DOES

Answers whether an agent is registered, whether it is on, when it really runs and what it has done. It keeps three facts apart: not registered, disabled and no recorded runs. Before anyone presses discover, it lists the rows discover would create and whether each would arrive enabled.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_registry_read_surface` A question about a name with no ai_agents row returns 'not registered', never 'disabled' and never 'zero runs': those are three different facts and the surface must not merge them.	Verified 27 Oct 2026	b461807
`platform.CAP.agent_discovery_sync` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-

NOT ALLOWED YET

- Move past answering questions. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Press discover, create a registry row, or switch an agent on or off.

Write-up: docs/capabilities/agent-registry-read-surface.md · Catalog row 17 · github.com/qninying/platform-reliability-agent

Cron run instrumentation

First release R2 | Ceiling R2 | Approved level on record: none

WHAT IT DOES

Answers whether a job ran, when, for how long and whether it failed, from what the cron wrappers recorded. For a timer the wrappers never see, the answer is 'not instrumented, no record exists', never 'did not run'.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.cron_run_instrumentation` For an untracked timer the answer is 'not instrumented: no record exists', never 'did not run'. A positive control: asking about a name known to have no ai_agents row must return the not-instrumented answer.	Verified 27 Oct 2026	b1eba88

NOT ALLOWED YET

- Move past answering questions. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Add instrumentation to the platform. It reads the records; it never becomes them.

Cron health alerting

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Names the cron jobs that are error-spiking or overdue, using the live check's own thresholds and selection. Jobs the check filters out are listed as not evaluated, so silence is never read as health.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.cron_health_alerting` Jobs excluded by the trigger_type and enabled filter are listed as not-evaluated rather than omitted, so the answer discloses its own blind spot.	Verified 28 Oct 2026	be1836d

NOT ALLOWED YET

- Run in shadow mode (R5). That waits on K.D.'s acceptable misses and false alarms.
- Raise the alert itself. It reproduces what the platform's check would raise.

NEVER, AT ANY LEVEL

- Change an alert threshold. That is K.D.'s decision.

Write-up: docs/capabilities/cron-health-alerting.md · Catalog row 19 · github.com/qninying/platform-reliability-agent

Agent health scoring

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Ranks agents worst first, scored the way the registry tab scores them, and labels any score built on zero runs as unscored rather than 100. At R4 it verifies a corrected scoring model on a sandbox copy.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX	SANDBOX			

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_health_scoring` Every score derived from run_count 0 is labelled unscored rather than 100.	Verified 28 Oct 2026	`10039fe`
`R4 sandbox rung` a corrected scoring model, verified in a sandbox	Built	`f7724ca`

NOT ALLOWED YET

- Put the corrected model into the live registry tab. That is a deploy a person performs.

NEVER, AT ANY LEVEL

- Change a live score or a live agent.

Write-up: docs/capabilities/agent-health-scoring.md · Catalog row 20 · github.com/qninying/platform-reliability-agent

Timer and agent registry reconciliation

First release R2 · Sandbox rung R4 built · Ceiling R4 · Approved level on record: none

WHAT IT DOES

Measures both lists from the platform's source at run time: untracked timers, phantom registry rows and schedules that disagree. Every count is dated and carries its names. For each untracked timer it proposes the row that would make it governable and says whether that row would arrive enabled.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX	SANDBOX			

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.registry_runtime_reconciliation` Counts are reported as the reconciler measured them with the measurement date attached; no count is restated from a document.	**Verified** 28 Oct 2026	`04dd0fb`
`platform.CAP.untracked_timer_registration_proposal` Every entry carries a real call-site citation and a measurement date; the published count is the one just measured, not 33 or 34 quoted from a document.	**Verified** 5 Nov 2026	`a15090c`
`R4 sandbox rung` the ratchet fails in both directions	**Built**	`9da41aa`
`R3 sandbox rung` proposed rows applied to a sandbox registry	**Built**	`a15090c`
`platform.CAP.runtime_cron_drift_detection` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-

NOT ALLOWED YET

- Apply a proposed row to the live registry. The R3 rehearsal applies them to a sandbox registry only.

NEVER, AT ANY LEVEL

- Create or change a registry row on the platform, or restate a count it did not just measure.

Write-up: docs/capabilities/registry-runtime-reconciliation.md · Catalog row 21 · github.com/qninying/platform-reliability-agent

Incident trace and error triage

First release R2 · Ceiling R5 · Approved level on record: none

WHAT IT DOES

Follows a failure from what a person sees back to the run that caused it, from any of four handles, and states the probable cause in the run's own words. Where no trace exists it says the job is untracked.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.incident_trace_and_error_triage` Where no trace exists for a failure, the answer is 'not traceable: this job is untracked', never a reconstructed narrative.	Verified 30 Oct 2026	`e9d8ecc`

NOT ALLOWED YET

- Run in shadow mode (R5) or open tickets on its own.

NEVER, AT ANY LEVEL

- Reconstruct a trace that was never recorded, or change the records it reads.

Write-up: docs/capabilities/incident-trace-and-error-triage.md · Catalog row 22 · github.com/qninying/platform-reliability-agent

Error classification coverage

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Reports what share of recorded failures carry one of the six standard error classes K.D. kept on 8 November, and names every catch site on the cron path that records none.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX	SANDBOX			

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.error_class_coverage` Coverage is reported as a share of all recorded failures; failures whose class is the generic fallback count as unclassified, not classified.	Verified 31 Oct 2026	`6a1c0af`
`R4 sandbox rung` classifies the cron path in a sandbox, grouped by kind	Built	`2c0d58b`

NOT ALLOWED YET

- Change a catch site in the platform. The R4 classification runs on a sandbox copy.

NEVER, AT ANY LEVEL

- Write to the platform's code.

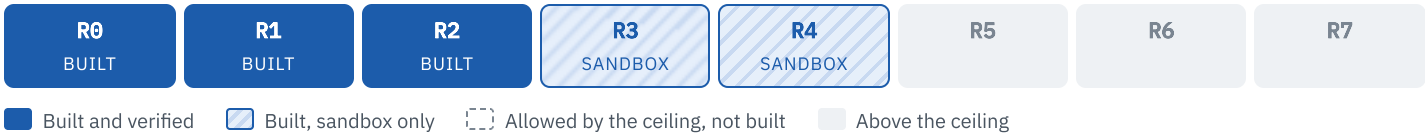
Write-up: docs/capabilities/error-class-coverage.md · Catalog row 26 · github.com/qninying/platform-reliability-agent

Scheduler registration liveness probe

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Names the one flag that gates every timer and the one function that starts the AI Ops scheduler, and says whether the platform's health route could ever fail when nothing registered. Given a boot log, it compares the timers registered with the 114 the code registers, and alarms on any shortfall.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.scheduler_registration_liveness_probe` The report names ENABLE_FOLLOWUP_SCHEDULER as the single flag and names startScheduler as the only caller of the AI Ops scheduler, both as measured facts.	Verified 2 Nov 2026	`b336b03`
`R4 sandbox rung` the probe, shown to fail with the flag unset	Built	`ae2ae55`

NOT ALLOWED YET

- Install the probe on the live platform. That is a deploy a person performs.

NEVER, AT ANY LEVEL

- Set the scheduler flag or restart the platform.

Write-up: docs/capabilities/scheduler-registration-liveness-probe.md · Catalog row 38 · github.com/qninying/platform-reliability-agent

Static security posture scan

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Reports what the platform's three security scanners (secrets, code, dependencies) last found and when. A scanner that has never run returns unknown, not 'no findings'.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.security_posture_scan` A scan that has never run returns 'unknown' rather than 'no findings'.	Verified 26 Oct 2026	8ad3ba9

NOT ALLOWED YET

- Run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Run a scan, fix a finding, or repeat a secret it has read.

Runtime threat and access watch

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Answers who holds which admin section today and what changed since the last sweep. It keeps 'no admin account' apart from 'an account with no sections', because they need different fixes.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.runtime_threat_and_access_watch` The answer distinguishes 'no admin_users row' from 'admin_users row with no sections', because they have different remedies.	Verified 26 Oct 2026	8263fa2

NOT ALLOWED YET

- Run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Grant, remove or create anyone's access. That is a privilege change.

Write-up: docs/capabilities/runtime-threat-and-access-watch.md · Catalog row 4 · github.com/qninying/platform-reliability-agent

Agent behaviour and AI-safety watch

First release R2 | Ceiling R6 | Approved level on record: none

WHAT IT DOES

Names the agents that acted outside their grants. Every finding names the specific grant that was missing; a finding that cannot name one is not emitted.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_behavior_safety_watch` Every finding names the specific grant that was missing; a finding with no named grant is not emitted.	Verified 26 Oct 2026	`f010814`

NOT ALLOWED YET

- Run in shadow mode (R5) or in a supervised production pilot (R6). Each needs its own approval.

NEVER, AT ANY LEVEL

- Stop or restrict an agent. Stopping a live timer stays a human decision.

Write-up: docs/capabilities/agent-behavior-safety-watch.md · Catalog row 5 · github.com/qninying/platform-reliability-agent

Security oversight rollup

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Gives one verdict over all seven security scanners, each listed as clean, dirty or unknown with its records. The verdict can never be green while any scanner is unknown.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.security_oversight_rollup` A scanner with no last_run_at renders as unknown; the rollup can never be green while any contribution is unknown.	Verified 26 Oct 2026	`a520ec2`

NOT ALLOWED YET

- Run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Change, start or silence a scanner.

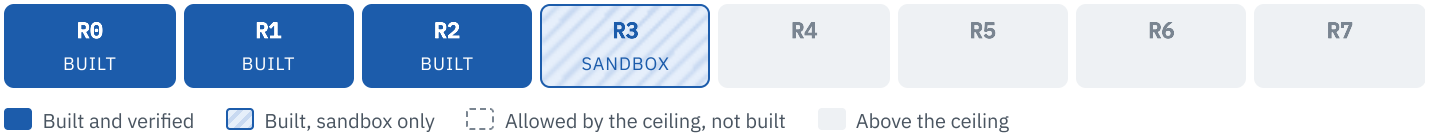
Write-up: docs/capabilities/security-oversight-rollup.md · Catalog row 6 · github.com/qninying/platform-reliability-agent

Backup and restore verification

First release R2 | Sandbox rung R3 built | Ceiling R3 | Human decision at every level | Approved level on record: none

WHAT IT DOES

States the recovery position and shows the search behind it. Today that position is unknown: no backup verification exists in the platform repository. At R3 it restores a supplied backup into a target it has first proven isolated, and only with a named operator.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.backup_and_restore_verification` The answer is 'unknown: no backup verification exists in this repository', with the search that establishes it; it never infers a position from the presence of pg_dump in unrelated scripts.	Verified 1 Nov 2026	`b59f0df`
`R3 sandbox rung` restores a supplied backup into an isolated target	Built	`1e16163`

NOT ALLOWED YET

- Restore anything unattended, or into a target it has not proven isolated.

NEVER, AT ANY LEVEL

- Take, read or restore a production backup. Production database access is denied to agents.

Write-up: docs/capabilities/backup-and-restore-verification.md · Catalog row 30 · github.com/qninying/platform-reliability-agent

Deploy verification evidence

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Judges one release's evidence: do the deployed commit, the running image and the live routes agree? A release missing any of the three is unverified, never deployed.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.deploy_verification_evidence` A release whose evidence is missing any of the three is reported as unverified, never as deployed; and an edge-level 200 alone is never accepted as proof the application started.	Verified 1 Nov 2026	`709e5b3`

NOT ALLOWED YET

- Move past judging evidence. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Reach the host, run a deploy, or decide whether a release stands.

Write-up: docs/capabilities/deploy-verification-evidence.md · Catalog row 31 · github.com/qninying/platform-reliability-agent

Orchestration auto-repair (sandbox only)

First release R4 · Ceiling R6 · Approved level on record: none

WHAT IT DOES

Runs the platform's two approved content repairs against a sandbox fixture, and proves the property that matters for anything on a timer: running the same repair twice leaves one change, not two.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	BUILT	BUILT			

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.orchestration_auto_repair` Running the same repair twice against the same fixture leaves one change, not two, and the second run reports no-op.	Verified 26 Oct 2026	add84c9

NOT ALLOWED YET

- Run in shadow mode (R5) or a supervised production pilot (R6). Each needs its own approval.

NEVER, AT ANY LEVEL

- Repair live course content.

Data retention sweeps

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Gives each table's retention window and when it was last swept. A window recorded nowhere is unknown, never a guessed number of days. At R4 it runs the sweep end to end on a sandbox copy.



Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.data_retention_sweep` A table whose window is not recorded anywhere returns 'unknown', never a guessed number of days.	Verified 26 Oct 2026	d267672
`R4 sandbox rung` the sweep, end to end in a sandbox	Built	3612aca

NOT ALLOWED YET

- Sweep live tables. Production deletes stay with the platform's own timers or a person.

NEVER, AT ANY LEVEL

- Delete production data. Deletion is irreversible.

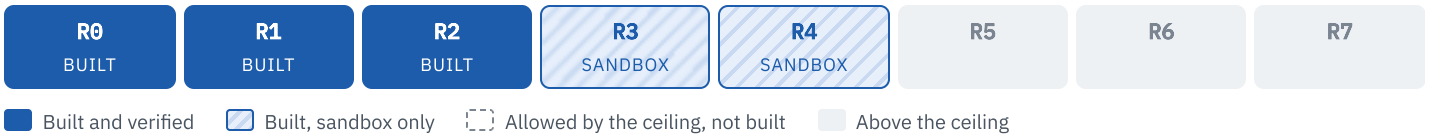
Write-up: docs/capabilities/data-retention-sweep.md · Catalog row 7 · github.com/qninying/platform-reliability-agent

Preview stack reaping

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Lists live preview stacks, how idle each is, and which are reap candidates. A stack with no activity signal is unknown-idle, not idle. At R4 it shows an interrupted teardown can be re-run and end clean, in a sandbox.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.preview_stack_reaping` A stack with no activity signal at all is reported as unknown-idle, not as idle.	**Verified** 26 Oct 2026	76a41d3
`R4 sandbox rung` an interrupted teardown, re-run, ends clean	**Built**	b519ce6

NOT ALLOWED YET

- Stop a live stack. Cloud resource deletion is denied to agents; the platform's own timer does it.

NEVER, AT ANY LEVEL

- Destroy running resources.

Write-up: docs/capabilities/preview-stack-reaping.md · Catalog row 8 · github.com/qninying/platform-reliability-agent

Stale action recovery

First release R2 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Says what is stuck, for how long and on which attempt, keeping 'past the threshold' apart from 'past the attempt cap', which must not be recycled. At R4 it runs the platform's recovery against a sandbox queue.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX	SANDBOX			

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.stale_action_recovery` The answer separates 'past the threshold' from 'past the attempt cap', because the second must not be recycled.	Verified 26 Oct 2026	`5ca2e3e`
`R4 sandbox rung` the platform's recovery, run against a sandbox queue	Built	`c97bff6`

NOT ALLOWED YET

- Recover a live action. The platform's own timer does that.

NEVER, AT ANY LEVEL

- Change live work state.

Write-up: docs/capabilities/stale-action-recovery.md · Catalog row 9 · github.com/qninying/platform-reliability-agent

Ticket auto-resolution (sandbox only)

First release R4 · Ceiling R7 · Approved level on record: none

WHAT IT DOES

Runs the platform's five ticket resolvers against a sandbox queue holding tickets that should close and tickets that must not, then runs all five again. Every must-not ticket stays open, and the second run closes nothing more.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	BUILT	BUILT			

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_ticket_auto_resolution` Every must-not ticket is still open, and a second run closes zero additional tickets and adds zero duplicate comments.	Verified 27 Oct 2026	`a2a40ee`

NOT ALLOWED YET

- Close a live ticket. Each resolver would need its own approval for R5 and above.

NEVER, AT ANY LEVEL

- Close live tickets without that approval.

Write-up: docs/capabilities/agent-ticket-auto-resolution.md · Catalog row 10 · github.com/qninying/platform-reliability-agent

Bug fix dispatch

First release R1 | Sandbox rung R4 built | Ceiling R4 | Human decision at every level | Approved level on record: none

WHAT IT DOES

For a reported bug, prepares an isolated reproduction and a baseline test that fails for the reported reason before any fix. At R4 it checks a proposed fix against that baseline in a sandbox with no network.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	SANDBOX	SANDBOX	SANDBOX			

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.platform_bug_fix_dispatch` The baseline test fails for the reported reason before any fix and is shown to fail: a test that passes before the fix proves nothing.	Verified 27 Oct 2026	872a586
`R4 sandbox rung` a proposed fix, verified in a sandbox	Built	b811e9a

NOT ALLOWED YET

- Write the fix itself, or let a sandbox test reach the network or other programs.

NEVER, AT ANY LEVEL

- Fix, merge or deploy platform code.

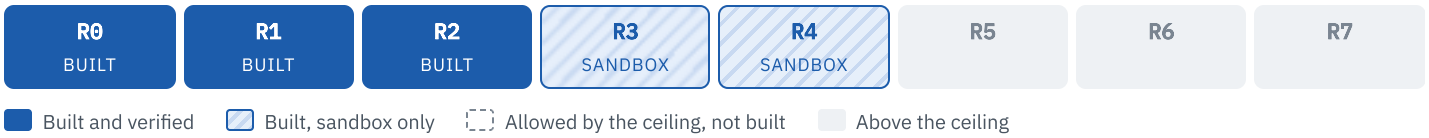
Write-up: docs/capabilities/platform-bug-fix-dispatch.md · Catalog row 16 · github.com/qninying/platform-reliability-agent

Dead-letter capture

First release R2 · Sandbox rung R4 built · Ceiling R4 · Approved level on record: none

WHAT IT DOES

Reports what has been dead-lettered, by error class, and names the job families that can never reach the dead-letter store, so an empty queue is never read as a healthy fleet.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.dead_letter_capture` The report names the unwrapped job families explicitly, so an empty queue is never read as a healthy fleet.	Verified 30 Oct 2026	`730bfae`
`R4 sandbox rung` the capture path, exercised across a restart	Built	`a8f21d9`

NOT ALLOWED YET

- Wire capture into the live platform. The R4 rehearsal runs on a sandbox copy.

NEVER, AT ANY LEVEL

- Write to the dead-letter queue.

Write-up: docs/capabilities/dead-letter-capture.md · Catalog row 24 · github.com/qninying/platform-reliability-agent

Dead-letter triage

First release R2 · Sandbox rung R4 built · Ceiling R4 · Human decision at every level · Approved level on record: none

WHAT IT DOES

Puts both dead-letter stores into one queue, grouped by error class, and proposes what a person should do with each entry. At R4 it verifies a replay end to end for publishing jobs in a sandbox.



Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.dead_letter_triage` An empty queue is reported as 'empty, and here is what could not have reached it', never as a healthy fleet.	**Verified** 30 Oct 2026	`ce04db2`
`R4 sandbox rung` a replay, verified end to end for publishing jobs	**Built**	`de98ddc`

NOT ALLOWED YET

- Replay a live entry. Replay is labelled unproven outside the sandbox.

NEVER, AT ANY LEVEL

- Replay, abandon or mark resolved any entry. A person decides at every level.

Write-up: docs/capabilities/dead-letter-triage.md · Catalog row 25 · github.com/qninying/platform-reliability-agent

Idempotency and replay audit

First release R1 | Sandbox rung R4 built | Ceiling R4 | Approved level on record: none

WHAT IT DOES

Runs a job twice in a sandbox and tells a job that is safe to run twice from one that is not. The harness is first shown to catch a job built to be unsafe. Every platform job stays unknown until it has been run twice.



Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.idempotency_and_replay_audit` The harness detects a deliberately non-idempotent fixture job as unsafe: if it cannot fail, it cannot pass.	Verified 31 Oct 2026	`ff99053`
`R4 sandbox rung` a twice-run verdict per job across five families	Built	`7d18457`

NOT ALLOWED YET

- Judge a job it has not run twice in a sandbox.

NEVER, AT ANY LEVEL

- Run a platform job twice in production.

Write-up: docs/capabilities/idempotency-and-replay-audit.md · Catalog row 27 · github.com/qninying/platform-reliability-agent

Cron overlap guard

First release R2 · Sandbox rung R4 built · Ceiling R4 · Approved level on record: none

WHAT IT DOES

Names every timer whose recorded run time can exceed its own interval, with both figures, and says which jobs hold their own claim. At R4 it builds a guard and proves it on an overrunning timer in a sandbox.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT	SANDBOX	SANDBOX			

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.cron_overlap_guard` Each name is accompanied by its interval and its recorded duration; a timer with no duration data is listed as unassessable rather than safe.	Verified 1 Nov 2026	`ece5f6c`
`R4 sandbox rung` the guard, proven on an overrunning timer	Built	`61f68b1`

NOT ALLOWED YET

- Apply the guard to a live timer. That is a deploy a person performs.

NEVER, AT ANY LEVEL

- Change a live timer.

Write-up: docs/capabilities/cron-overlap-guard.md · Catalog row 28 · github.com/qninying/platform-reliability-agent

Production deploy review

First release R1 · R1 is final · Human decision at every level · Approved level on record: none

WHAT IT DOES

Reports, as measured facts, whether a checkout is clean and whether it matches origin/main. It acts on none of them. R1 is where this capability stops for good.

R0 BUILT	R1 BUILT	R2	R3	R4	R5	R6	R7

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.production_deploy_review_and_approval` The readiness report states tree cleanliness and whether HEAD matches origin/main as measured facts; it never merges, pushes to main or deploys, and a test asserts no deploy tool is among its grants.	Verified 1 Nov 2026	`db13ddf`

NOT ALLOWED YET

- Anything beyond R1. No rung can grant more.

NEVER, AT ANY LEVEL

- Approve, merge or deploy. Both are denied to agents by the platform.

Write-up: docs/capabilities/deploy-readiness.md · Catalog row 32 · github.com/qninying/platform-reliability-agent

Infrastructure evolution proposals

First release R2 | Ceiling R5 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Answers what the platform's three architect agents have proposed, why, and what happened to each: open, accepted, deferred or rejected.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.infrastructure_evolution_proposals` Each proposal's current state is one of open, accepted, deferred or rejected; a proposal with no recorded outcome reads as open, not as accepted.	Verified 27 Oct 2026	`20f265e`

NOT ALLOWED YET

- Run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Accept, defer or reject a proposal. That is a human decision at every level.

Write-up: docs/capabilities/infrastructure-evolution-proposals.md · Catalog row 11 · github.com/qninying/platform-reliability-agent

Agent enable, disable and pause control

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Says whether an agent should be stopped, why, what stopping it would actually stop, and whether pause or disable would hold. For halting the whole fleet, it says which lever reaches what and what resuming would undo.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_enable_disable_control` A recommendation for a shared agent_name lists all timers that share it; a recommendation for an unregistered name states that no disable flag exists for it rather than proposing one.	**Verified** 27 Oct 2026	2cccba9
`platform.CAP.emergency_stop_and_kill_switch` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-

NOT ALLOWED YET

- Move past recommending. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Stop, start, pause or resume a live agent, or call the control route. A person acts.

Write-up: docs/capabilities/agent-enable-disable-control.md · Catalog row 13 · github.com/qninying/platform-reliability-agent

Workforce technology direction

First release R2 · Ceiling R5 · Human decision at every level · Approved level on record: none

WHAT IT DOES

Lists the tooling the fleet holds and where it diverges from the written direction. With no written direction on file, it says so rather than inventing one.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.workforce_technology_direction` Divergence is reported against a written intended direction; with no written direction on file, the answer is 'unknown', not 'no divergence'.	Verified 27 Oct 2026	22cff65

NOT ALLOWED YET

- Run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Set the direction, or change the AI model class. A model change is an escalation for a named person.

Write-up: docs/capabilities/workforce-technology-direction.md · Catalog row 14 · github.com/qninying/platform-reliability-agent

Prompt and performance optimisation

First release R2 · Ceiling R5 · Human decision at every level · Approved level on record: none

WHAT IT DOES

Ranks prompts by cost and latency from recorded runs, and lists open change proposals. A prompt with no recorded runs has an unknown cost, not $0.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified · Built, sandbox only · Allowed by the ceiling, not built · Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.prompt_and_performance_optimization` Every cost or latency figure traces to recorded runs; a prompt with no recorded runs reports unknown rather than zero.	**Verified** 27 Oct 2026	`67e2bc0`

NOT ALLOWED YET

- Propose a prompt change of its own, or run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Adopt a prompt change. That changes production behaviour; K.D. decides.

Write-up: docs/capabilities/prompt-and-performance-optimization.md · Catalog row 15 · github.com/qninying/platform-reliability-agent

Schedule governance and live reload

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Proposes schedule changes with the measured run cost and signal rate behind each, and says whether the timer can be reloaded live or only with a restart, because that decides the remedy.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.schedule_governance_live_reload` Every proposal says whether the target is in the reloadable aiOps registry or inline in schedulerService.ts, because the two have different remedies; a proposal with an unknown home is labelled unknown.	Verified 30 Oct 2026	1052ddd

NOT ALLOWED YET

- Move past proposing. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Change or switch off a live timer.

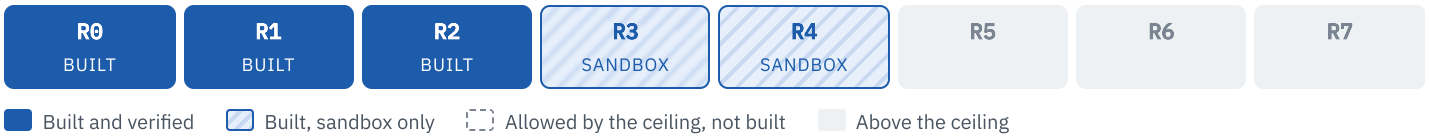
Write-up: docs/capabilities/schedule-governance-and-live-reload.md · Catalog row 23 · github.com/qninying/platform-reliability-agent

Default-on timer guard

First release R2 | Sandbox rung R4 built | Ceiling R4 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Names every scheduler timer that has no governing row and so runs by default, measured from the source each run, and cites the code that applies that default. At R4 it builds a guard and proves it fails closed in a sandbox.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.default_on_timer_guard` The enumeration is a measured list of names, not a count; and it states the three code locations where the default is applied.	Verified 2 Nov 2026	279489b
`R4 sandbox rung` the guard, proven to fail closed in a sandbox	Built	af47401

NOT ALLOWED YET

- Apply the guard on the live platform. That is a deploy a person performs.

NEVER, AT ANY LEVEL

- Decide whether an ungoverned timer should run, or switch the default off. Both are K.D.'s.

Write-up: docs/capabilities/default-on-timer-guard.md · Catalog row 37 · github.com/qninying/platform-reliability-agent

Reliability briefing

First release R2 | Ceiling R5 | Approved level on record: none

WHAT IT DOES

Produces the manager's briefing on request: what ran, what failed, what nobody can measure, and which decisions need a person. No percentage appears without the record count behind it.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.reliability_briefing_for_manager` No percentage appears without the record count behind it; the unmeasured section names the untracked timers and the phantom crons as present rather than omitting them.	Verified 1 Nov 2026	bd6336c

NOT ALLOWED YET

- Run on a schedule or in shadow mode (R5).

NEVER, AT ANY LEVEL

- Send the briefing. Live email is denied to agents; the platform's briefing service delivers it.

Write-up: docs/capabilities/reliability-report.md · Catalog row 33 · github.com/qninying/platform-reliability-agent

Reliability threshold ownership

First release R2 | Ceiling R5 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Puts every number that decides when the platform complains in one list, with its file and line, how its alerts are delivered and what it is producing. A threshold with no recorded outcomes is unmeasured, never well-tuned.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.reliability_threshold_ownership` Each threshold is reported with its source file and line; a threshold whose alert outcomes were never recorded is reported as unmeasured rather than as well-tuned.	Verified 1 Nov 2026	`efff59b`

NOT ALLOWED YET

- Propose a value, or run in shadow mode (R5).

NEVER, AT ANY LEVEL

- Set a threshold. That is K.D.'s decision at every level.

Write-up: docs/capabilities/reliability-threshold-ownership.md · Catalog row 34 · github.com/qninying/platform-reliability-agent

Agent promotion evidence

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Assembles the evidence for one capability at one rung and states plainly whether it meets that rung's criteria. Any criterion with no evidence is named as unmet.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.agent_promotion_and_approval_decision` The assembly cites real run records and names any criterion with no evidence as unmet; a capability whose only evidence is a passing mocked test is reported as unproven.	Verified 1 Nov 2026	02842d9

NOT ALLOWED YET

- Move past assembling evidence. R2 is this capability's ceiling, permanently.

NEVER, AT ANY LEVEL

- Grant a rung, record a decision, or promote itself. An agent that could would not be governed.

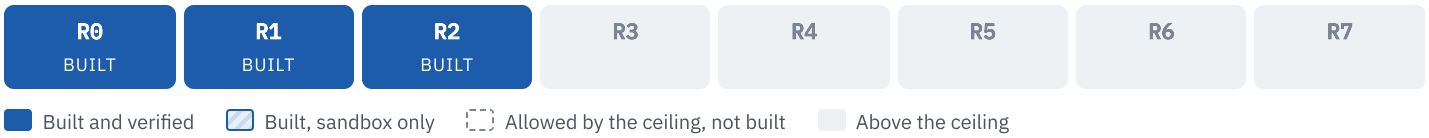
Write-up: docs/capabilities/agent-promotion-and-approval-decision.md · Catalog row 35 · github.com/qninying/platform-reliability-agent

Autonomy authority resolution

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

Answers what a capability's effective authority is today and which constraint binds it. No recorded approval resolves to none, never to a defaulted rung. Today all 48 capabilities resolve to none.



EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.capability_autonomy_authority_resolution` A capability with no recorded approval returns null with reason no_approved_autonomy, never a defaulted rung; and a human_decision capability never resolves above autonomy:R5.	Verified 2 Nov 2026	`a00c719`

NOT ALLOWED YET

- Move past answering. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Raise any capability's level, its own included.

Write-up: docs/capabilities/capability-autonomy-authority-resolution.md · Catalog row 36 · github.com/qninying/platform-reliability-agent

Manager access

First release R2 | Ceiling R2 | Human decision at every level | Approved level on record: none

WHAT IT DOES

For one agent, answers who may act on it and by which route, whether its accountable manager can reach it, and which directives actually take effect. A directive with no enforcement path is recorded but not in force.

R0	R1	R2	R3	R4	R5	R6	R7
BUILT	BUILT	BUILT					

Built and verified | Built, sandbox only | Allowed by the ceiling, not built | Above the ceiling

EVIDENCE

STORY AND ITS ACCEPTANCE CRITERION	STATE	COMMIT
`platform.CAP.manager_scoped_authorization` For K.D. and any platform agent the answer is 'nobody, via no route, because he holds no admin_users row': a positive control that the gap is reported rather than papered over.	**Verified** 4 Nov 2026	`cdec00f`
`platform.CAP.manager_reachable_agent_surface` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-
`platform.CAP.manager_restriction_directives` Folded into this capability under A.M.'s ruling of 4 November; its criterion is answered here.	Folded	-

NOT ALLOWED YET

- Move past answering. R2 is this capability's ceiling.

NEVER, AT ANY LEVEL

- Grant a login, a management role or a manager.

Write-up: docs/capabilities/manager-access.md · Catalog row 39 · github.com/qninying/platform-reliability-agent

The rest of the plan, and what it waits on

Of the 48 capabilities in the platform's plan, 40 stories are verified and mapped above. Six more are folded or merged into a built capability under A.M.'s ruling of 4 November, and appear on that capability's page. Two are held.

HELD

CAPABILITY	WHY
Agent identity attribution `platform.CAP.agent_identity_attribution`	Held until the governance gate and tool broker exist (A.M., 4 November).
Execution policy enforcement `platform.CAP.execution_policy_enforcement`	Held until the governance gate and tool broker exist (A.M., 4 November).

WHAT MOVES ELLIS UP THE LADDER

Shadow mode (R5)	The monitoring capabilities go next to shadow mode, compared against approved reference outcomes. That needs K.D.'s severity rules and acceptable misses and false alarms. Nothing on this map is blocked on them.
An approved level	Each step up is a recorded human decision: K.D. to R2, A.M. from R3. Ellis assembles the evidence (Agent promotion evidence) and resolves the result (Autonomy authority resolution); it never grants a level.
Gate and broker	The governance gate and tool broker turn these checks into tools Ellis calls under policy. The two held capabilities wait on them.

Sources: .colaberry/progress.json (the platform's verification record), docs/capabilities/*.md, README autonomy table. Verified state read on 10 November 2026.